

Problem #2

```
A=zeros(1,10);Bx=zeros(1,10);C=zeros(1,10);
D=zeros(1,10);E=zeros(1,10);
thetab=asind(3/5)+270; %angle for b in degrees
y = [0 0 -74*10^3 0 24*10^3 0 0 0 0 0];
y=y';
```

```
Ax=zeros(1,10);
Ax(1,2)=1;Ax(1,8)=1;
```

```
Ay=zeros(1,10);
Ay(1,1)=1; Ay(1,9)=1;
```

```
Bx = zeros(1,10); % force in x of B
Bx(1,4)=1;Bx(1,3)=1*cosd(thetab);
By=zeros(1,10);% force in y of B
By(1,1)=-1;By(1,3)=1*sind(thetab);
```

```
thetac = thetab;
Cx=zeros(1,10); %Cx force
Cx(1,4)=-1;Cx(1,7)=cosd(thetac);
Cy=zeros(1,10); % Cy force
Cy(1,5)=-1; Cy(1,7)=sind(thetac);
```

```
thetad=90+asind(3/5);
Dx=zeros(1,10);%Dx force
Dx(1,2)=-1;Dx(1,6)=1;Dx(1,3)=cosd(thetad);
Dy=zeros(1,10); %Dy force
Dy(1,5)=1;Dy(1,3)=sind(thetad);
```

```
thetae=180-thetac;
Ex=zeros(1,10); % Ex
Ex(1,6)=-1;Ex(1,7)=cosd(thetae);
Ey=zeros(1,10);
Ey(1,10)=1;Ey(1,7)=sind(thetae);
```

```
Matr=[Ax;Ay;Bx;By;Cx;Cy;Dx;Dy;Ex;Ey];
sort(abs(Matr\y))
```

```
ans = 10x1
10^4 ×
2.5000000000000001
3.3333333333333330
3.3333333333333330
3.3333333333333331
3.3333333333333331
4.1666666666666663
4.1666666666666665
4.9000000000000000
4.9999999999999998
4.9999999999999998
```

Note: this is my first time doing a truss analysis since I dont have a background in civil engineering.